



Course

Cs-551(ARTIFICIAL INTELLIGENCE)

Project

OBJECT DETECTION SYSTEM

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1. Objective

The objective of this project is to build a system that can automatically detect vehicle license plates in an image or webcam feed. This is done by training a YOLO (You Only Look Once) deep learning model on a custom dataset of license plate images, then wrapping the trained model in a simple web interface so it can be tested and demonstrated easily.

2. Introduction

License plate detection is a common real-world application of computer vision, used in traffic monitoring, toll systems, and parking management. Instead of using a generic object detector, this project fine-tunes a pretrained YOLO model specifically on license plate images, so it learns to recognize plates accurately rather than general objects like people or cars.

3. Tools and Technologies Used

- **Python** — programming language used for the whole project
- **Ultralytics YOLO** — pretrained model, fine-tuned on a custom dataset
- **Roboflow** — used to prepare and label the license plate dataset
- **Gradio** — used to build the web interface for testing the model
- **Pandas** — used to display detection results in a table

4. Methodology

The project was carried out in three main stages:

- **Dataset Preparation:** A license plate dataset was obtained and labeled using Roboflow, then exported in a YOLO-compatible format.
- **Model Training:** A pretrained YOLO model was fine-tuned on the license plate dataset so it could learn to recognize plates specifically, rather than general objects.
- **Interface Building:** The trained model was connected to a simple Gradio web interface, allowing a user to upload an image or use a webcam and instantly see detected license plates along with confidence scores.

5. Results

The trained model was tested on several images containing vehicles with visible license plates. The system successfully drew a bounding box around each detected plate and displayed a confidence score. Adjusting the confidence threshold slider showed the expected trade-off: lower values detected more plates (including some uncertain ones), while higher values showed only the most confident detections.

[Insert screenshots here: sample input image, annotated output with detected plate, and the web interface.]

6. Conclusion

This project demonstrates how a general-purpose object detection model can be fine-tuned for a specific real-world task using a custom dataset. The resulting system is able to detect license plates with reasonable accuracy and is presented through an easy-to-use web interface, making it simple to test and demonstrate.